

Bowen Feng

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Final-year MInf student at the University of Edinburgh working on human-AI collaboration: auditable workflows for LLM agents, evidence that travels with analysis, and interfaces for handoff and oversight.

EDUCATION

University of Edinburgh 2022 - 2027 (expected)

Master of Informatics (MInf), five-year integrated master's program, School of Informatics, Edinburgh, UK. Overall average 70/100.

Relevant coursework: Human-Computer Interaction, User Research and Interaction Design, Machine Learning, Deep Learning, Computer Vision.

University of California, Davis 2024 - 2025

Exchange student, Davis, CA, United States (GPA 3.82/4.0)

RESEARCH INTERESTS

Human-AI collaboration and human oversight of LLM and web agents; provenance, sensemaking and analytical handoffs; auditing of multimodal assistants; trust and verification interfaces.

SELECTED RESEARCH PROJECTS

Evidence Bundles for Analytical Handoffs

First-author research | Human-computer interaction

- Ran a controlled study with 12 data workers comparing chronological transcripts, artifact landmarks and context-preserving evidence bundles for producing technical and executive handoffs.
- Evidence bundles improved handoff fidelity and caveat retention and reduced unsupported claims relative to the other two interfaces; follow-up interviews explained the mechanisms.

TTAT: Auditable Web Workflows

First-author research with Prof. Tian Xie, Zhejiang University

- Built a six-stage system connecting human-authored task templates to bounded GUI actions, approved scripts, semantic analysis and reviewable evidence; 88.67% run success on the full 812-task WebArena suite.
- Evaluated with 12 industrial users against usual practice, measuring main-task completion, workflow lifecycle completion and audit judgment accuracy.

Auditing Visual Assistants under Workflow Obligations

First-author research | Multimodal systems

- Correct records can make a visual assistant answer about the wrong image region and then certify that answer as supported. Designed a counterfactual audit that varies records and workflow obligations while holding the visual query fixed, over 80 charts and 362 document pages, with semantic controls that separate coverage obligations from wording and record-field names.

PUBLICATIONS

Micro-Expression Recognition System Based on Dual-Stream Graph Structure and Action Unit Auxiliary Learning.
Bowen Feng (sole author).

International Conference on Intelligent Sensing, Communications and Marine Applications (ISCMIA 2026), accepted.

SPR: A Similar Projection Revisor for Complex Logical Reasoning over Knowledge Graphs.

Tingting Wang, Yuxuan Tang, Ruolin Li, Duo Yu, **Bowen Feng**, Feng Ding, Shuo Yu, and Yanming Shen.

Web Information Systems and Applications (WISA 2024), LNCS 14883, pp. 78-90. doi.org/10.1007/978-981-97-7707-5_7

Probabilistic Dual-Adaptive Spatio-Temporal Graph Convolutional Networks for Forecasting Energy Consumption Dynamics of Electric Vehicle Charging Stations.

Djamel Eddine Mekkaoui, Mohamed Amine Midoun, Abdelkarim Smaili, **Bowen Feng**, Mohamed Zakariya Talhaoui, and Yanming Shen.
Computers & Electrical Engineering, 122, 109976 (2025). doi.org/10.1016/j.compeleceng.2024.109976
TIDE-IS: Teacher-Aligned Instance Reliability and Bounded Contour Refinement for Underwater Instance Segmentation.
Bowen Feng (sole author).
International Conference on Intelligent Sensing, Communications and Marine Applications (ISCMA 2026), accepted.

UNDER REVIEW

From Artifacts to Evidence Bundles: How Context-Preserving Navigation Shapes Analytical Handoffs.
Bowen Feng. Under review, 2026.
From Task Templates to Action Traces: Auditable Web-Community Workflows for Industrial Users.
Bowen Feng and Tian Xie. Under review, 2026.

MANUSCRIPTS IN PREPARATION

When Correct Evidence Becomes a Target: Auditing Obligation-Induced Capture in Visual Workflows *First author. Manuscript in preparation, 2026.*
MOVE Before You Click: Visual Crossover for Agent Grounding *First author. Manuscript in preparation, 2026.*

RESEARCH EXPERIENCE

Research with Prof. Tian Xie, Zhejiang University 2025 - 2026
Auditable web workflows (TTAT): system design, WebArena evaluation and a 12-user study.
Research assistant, Prof. Yanming Shen's group, Dalian University of Technology 2022 - 2024
Knowledge-graph reasoning (WISA 2024) and spatio-temporal forecasting (*Computers & Electrical Engineering* 2025): data processing, temporal modeling components and baseline comparisons.
Bioinformatics and Statistical Analysis Intern, CAICT, Beijing 2025
Prepared data and covariates in R and Python; model selection, multiple-comparison control, effect sizes and confidence intervals; sensitivity analyses and reproducible reports.

SERVICE

Student Representative (Year 4), School of Informatics, University of Edinburgh 2025 - 2026
Programme Representative, Computer Science, School of Informatics, University of Edinburgh 2023 - 2024
Year 2 Student Representative, School of Informatics, University of Edinburgh 2023 - 2024

SKILLS

Programming: Python, R, C++, JavaScript, LaTeX.
Frameworks and tools: PyTorch, OpenCV, NumPy/SciPy, scikit-learn, pandas, Playwright and Selenium for browser automation, Git; training on multi-GPU clusters.
Methods: Controlled and counterfactual evaluation, paired-capture datasets, calibration and conformal prediction, ablation studies, user studies with statistical inference, failure analysis.